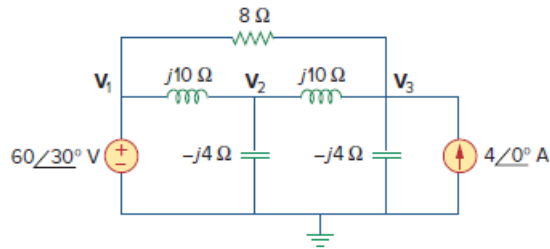


Problem

Use *PSpice* or *MultiSim* to find $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ in the network of Fig. 10.128.

Figure 10.128:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 10.128 in the textbook.

The node voltage $\mathbf{V}_1$ is,

$$\mathbf{V}_1 = 60\angle 30^\circ \text{ V}$$

Apply Kirchhoff's current law at node $\mathbf{V}_2$.

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{j10} + \frac{\mathbf{V}_2}{-j4} + \frac{\mathbf{V}_2 - \mathbf{V}_3}{j10} = 0$$

$$\left(\frac{1}{j10} + \frac{1}{-j4} + \frac{1}{j10} \right) \mathbf{V}_2 - \frac{\mathbf{V}_3}{j10} = \frac{\mathbf{V}_1}{j10}$$

$$\left(\frac{1}{10} + \frac{1}{-4} + \frac{1}{10} \right) \mathbf{V}_2 - \frac{\mathbf{V}_3}{10} = \frac{\mathbf{V}_1}{10}$$

$$-0.05\mathbf{V}_2 - 0.1\mathbf{V}_3 = 0.1\mathbf{V}_1$$

Substitute $60\angle 30^\circ \text{ V}$ for $\mathbf{V}_1$.

$$-(0.05)\mathbf{V}_2 - 0.1\mathbf{V}_3 = (0.1)(60\angle 30^\circ)$$

$$(0.5)\mathbf{V}_2 + \mathbf{V}_3 = -(60\angle 30^\circ)$$

$$0.5\mathbf{V}_2 + \mathbf{V}_3 = -51.96 - j30$$

$$\mathbf{V}_3 = -51.96 - j30 - 0.5\mathbf{V}_2 \dots\dots (1)$$

Step 2 of 7

Apply Kirchhoff's current law at node $\mathbf{V}_3$.

$$\frac{\mathbf{V}_3 - \mathbf{V}_1}{8} + \frac{\mathbf{V}_3}{-j4} + \frac{\mathbf{V}_3 - \mathbf{V}_2}{j10} = 4\angle 0^\circ$$

$$-\frac{\mathbf{V}_2}{j10} + \left(\frac{1}{8} + \frac{1}{-j4} + \frac{1}{j10} \right) \mathbf{V}_3 = 4\angle 0^\circ + \frac{\mathbf{V}_1}{8}$$

$$(j0.1)\mathbf{V}_2 + (0.125 + j0.15)\mathbf{V}_3 = 4\angle 0^\circ + 0.125\mathbf{V}_1$$

Substitute $60\angle 30^\circ \text{ V}$ for $\mathbf{V}_1$.

$$(j0.1)\mathbf{V}_2 + (0.125 + j0.15)\mathbf{V}_3 = 4\angle 0^\circ + 0.125(60\angle 30^\circ)$$

$$(j0.1)\mathbf{V}_2 + (0.125 + j0.15)\mathbf{V}_3 = 4 + 6.495 + j3.75$$

Substitute $-51.96 - j30 - 0.5\mathbf{V}_2$ for $\mathbf{V}_3$.

$$(j0.1)\mathbf{V}_2 + (0.125 + j0.15)(-51.96 - j30 - 0.5\mathbf{V}_2) = 10.495 + j3.75$$

$$(j0.1 - 0.0625 - j0.075)\mathbf{V}_2 - 1.995 - j11.54 = 10.495 + j3.75$$

$$(-0.0625 + j0.025)\mathbf{V}_2 = 12.49 + j15.29$$

$$\begin{aligned}\mathbf{V}_2 &= \frac{12.49 + j15.29}{-0.0625 + j0.025} \\ &= -87.917 - j279.8 \\ &= 293.29\angle -107.44^\circ \text{ V}\end{aligned}$$

Step 3 of 7

Recall equation (1).

$$\mathbf{V}_3 = -51.96 - j30 - 0.5\mathbf{V}_2$$

Substitute $293.29\angle -107.44^\circ \text{ V}$ for $\mathbf{V}_2$.

$$\mathbf{V}_3 = -51.96 - j30 - (0.5)(293.29\angle -107.44^\circ \text{ V})$$

$$= -51.96 - j30 - (0.5)(-87.917 - j279.8)$$

$$= -51.96 - j30 + 43.95 + j139.9$$

$$= -8.011 + j109.9$$

$$= 110.2\angle 94.17^\circ \text{ V}$$

Therefore, the node voltages are,

$\begin{aligned}\mathbf{V}_1 &= 60\angle 30^\circ \text{ V} \\ \mathbf{V}_2 &= 293.29\angle -107.44^\circ \text{ V} \\ \mathbf{V}_3 &= 110.2\angle 94.17^\circ \text{ V}\end{aligned}$
--

Step 4 of 7

PSpICE simulation:

Assume frequency of the sources are,

$$\omega = 1 \text{ rad/s}$$

Frequency in Hz is,

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{1}{2\pi} \\ &= 0.159155 \text{ Hz} \end{aligned}$$

Calculate the value of capacitor.

$$\begin{aligned} C &= \frac{1}{\omega X_C} \\ &= \frac{1}{(1)(4)} \\ &= 0.25 \text{ F} \end{aligned}$$

Calculate the value of inductor.

$$\begin{aligned} L &= \frac{X_L}{\omega} \\ &= \frac{10}{1} \\ &= 10 \text{ H} \end{aligned}$$

Step 5 of 7

PSpice simulation circuit is shown in Figure 1.

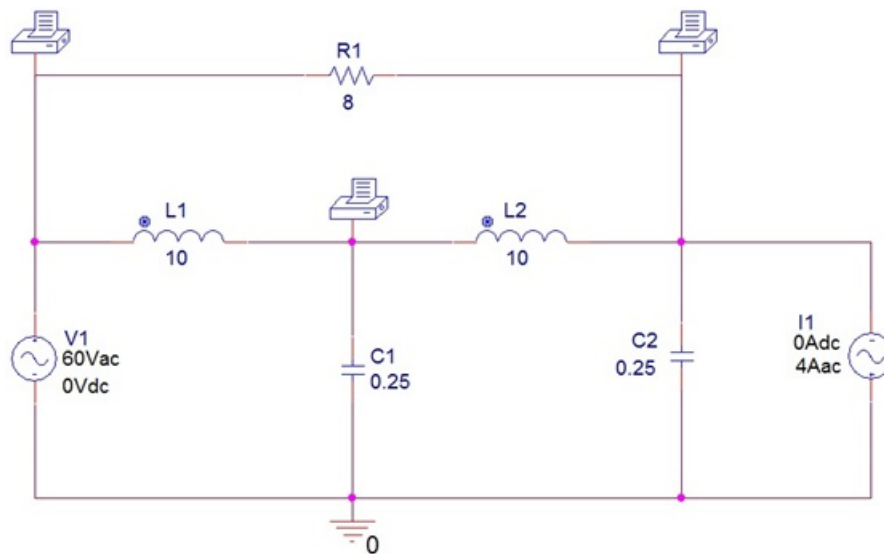


Figure 1

Step 6 of 7

Simulation setting window is shown in Figure 2.

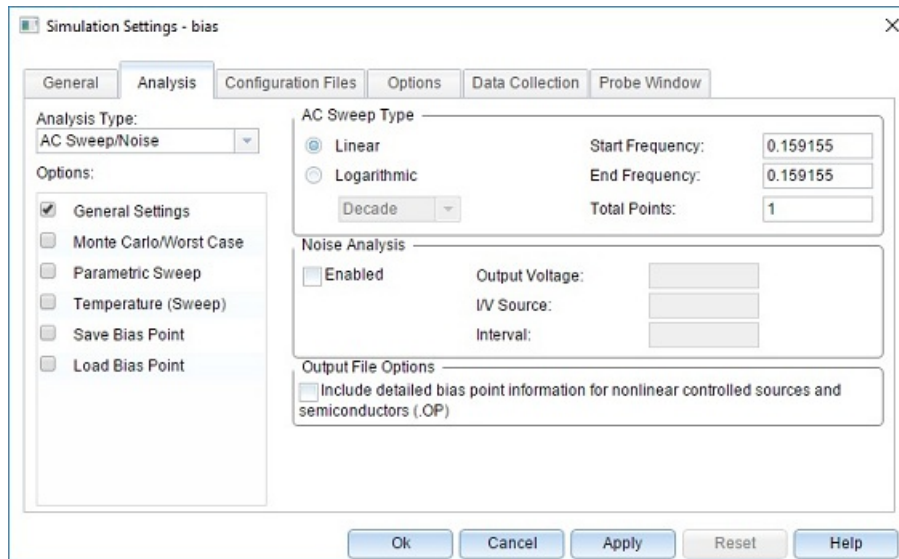


Figure 2

Step 7 of 7

The output file is shown in Figure 3.

```

FREQ      VM(N14477)  VP(N14477)  VR(N14477)  VI(N14477)
1.592E-01  6.000E+01  3.000E+01  5.196E+01  3.000E+01

FREQ      VM(N14558)  VP(N14558)  VR(N14558)  VI(N14558)
1.592E-01  2.933E+02 -1.074E+02 -8.790E+01 -2.799E+02

FREQ      VM(N14482)  VP(N14482)  VR(N14482)  VI(N14482)
1.592E-01  1.102E+02  9.417E+01 -8.011E+00  1.099E+02

JOB CONCLUDED

```

Figure 3

From the output file, the output voltages are,

$$V_1 = 60\angle 30^\circ \text{ V}$$

$$V_2 = 293.29\angle -107.44^\circ \text{ V}$$

$$V_2 = 110.2\angle 94.17^\circ \text{ V}$$

The voltages obtained in both cases (theoretically and Pspice simulation) are same.